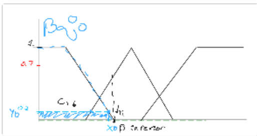
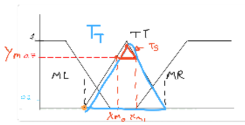
Centroides



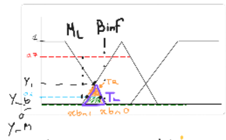
C_Sq_low = x_low/2

C_T_low = x_low+(Low.R-x_low)*2/3



TS = (x_mid1+ x_mid0+ Mid.M)/3

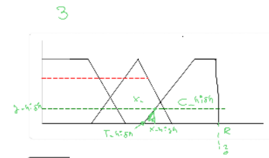
TT = (Mid.R+Mid.L+Mid.M)/3



T2 = (x_lm1+x_lm0+ X1) /3

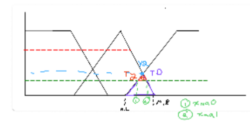
TL = (Low.R + Mid.L + X1)/3

A_inter1 = TL -T2



Sq_high = (R +x_high)/2

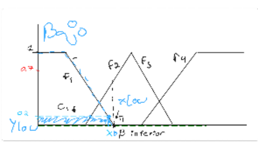
T_high = (x_high + High.L)/2



TR = (M.R+A.L+X2)/3

T2 = (x_mh1 +x_mh0+X2)/3

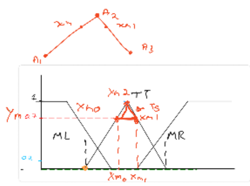
Areas



Sq_Low = y_Low*x_Low

T_LOW = y_Low*(Low.R -x_Low)/2

A_Low= T1 + C1

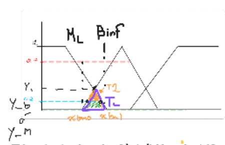


TS = (x_mid1- x_mid0) *(1-y_mid)/2

TT = (M.R-M.L)/2

A2 = TT - TS

$A_1 \cap A_2$



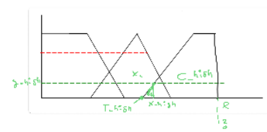
T1 = (x_lm1-x_lm0) * (Y1-y_lm)/2

TL = Y1 * (B.R - M.L)/2

A_inter1 = TL -T1

y_bm = min(y_b,y_m)
if(y_bm<Y1)
T2= (x_b1...)
else T2 = 0

A 3



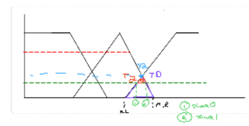
Sq_high = (R -x_high)*y_high

T_high = (x_high - A.L)*y_high/2

A3 = C_a + T_a

x_high= f_High(y_high)

$A_3 \cap A_2$



y_ma = min(y_m,y_a)
if(y_ma<Y2)
T3= (x_ma1...)
else T3 = 0

TR = (M.R-A.L)* Y2/2

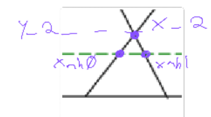
T2 = (x_mh1 -x_mh0)* (Y2 - y_mh)/2

A_inter2 = TD- T3

x_mh0 = f_High(y_mh)

x_mh1 = f_Mid2(y_mh)

X_2 = f_High(Y2) o f_Mid2(Y2)



AT = A1 +A2 + A3 - A_inter1 - A_inter 2

AT = Sq_b + T_b + TT - TS + Sq_a + T_a + T1 + T2 -TR - TL

y_low = -m_low * x_low + Low.R
(y_low-Low.r) = -m_low *x_low
x_low = (Low.r-y_low)/m_low
x_low =f_low(y_low)

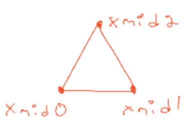
f_low(y) = (Low.R - y_low)/m_low
f_Mid0(y) = (y_mid - Mid.L)/m_Mid0
f_Mid1 = (Mid.R - y_mid)/m_Mid1
f_High =(y_high-High.L)/m_High

y=mx+C
y_m=x/(M.C-M.L) +C
(y_m-M_L)(M.C-M.L)=x_m0

y=mx+C
y_m=x/(M.C-M.L) +C, C =M.L
[y_m-M.L](M.C-M.L)=x_b0

x_mid0 = (y_mid - Mid.L)/m_Mid0
f_Mid1

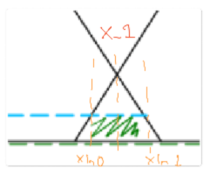
x_mid1 = (Mid.R - y_mid)/m_Mid1
f_Mid2

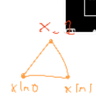


x_lm0 = (y_lm - Mid.L)/m_Mid0
f_Mid1

x_lm1 = (Low.R - y_lm)/m_low
f_low

X_1 = f_Mid1(Y1) o f_low(Y1)





$$x = \frac{(MM - MI)\,LS + (LS - LI)\,MI}{LS + MM - LI - MI}$$

y=mx+C
y_h=x/(H.R-H.L) +C, C =H.L
[y_h-H.L](H.R-H.L)=x_b0

(%i6) eq2 : (MR−x)/(MR−MM) = (x−HL)/(HR−HL);

(eq2) $\frac{MR-x}{MR-MM} = \frac{x-HL}{HR-HL}$

(%i7) solve(eq2,x);

(%o7) $\left[x = \frac{HR\,MR - HL\,MM}{MR - MM + HR - HL}\right]$

(%i2) eq1 : −(x−LS)/(LS−LI) = (x−MI)/(MM−MI)
;

(eq1) $\frac{LS-x}{LS-LI} = \frac{x-MI}{MM-MI}$

(%i3) solve(eq1,x);

(%o3) $\left[x = \frac{LS\,MM - LI\,MI}{MM - MI + LS - LI}\right]$